

Frais Asghar

Third Year Undergraduate

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SUMMARY:

Mechanical Engineering student with a strong focus on CAD, CFD, FEA, and thermal-fluid engineering, with hands on experience in designing and optimization of heat exchangers and flow systems through computational analysis. Experienced in SolidWorks and COMSOL Multiphysics, including simulation of Y Shaped inserts tube, Baffled Plate shell-and-tube, swirled-tube, and enhanced heat transfer configurations. Complemented by a 6 week CAD and simulation internship and proficiency in CAD and simulation, MATLAB, ANSYS, and Python, with a strong ability to translate engineering designs into quantified, simulation driven improvements.

EDUCATION

National University of Sciences & Technology (NUST) Bachelor of Engineering in Mechanical Engineering (4 th Semester)	2024-2028
City College of Science & Commerce F.Sc. Pre-Engineering (Percentage: 87.67)	2022-2024
Ansari Science High School Matriculation (Percentage: 98.45)	2020-2022

Research Experience:

Heat Transfer & CFD Research Intern:

Supervisor: *Syed Maaz Hasan, AIMS Lab, SINES, NUST*

• *June 2026 - August 2026*

- Investigated heat transfer enhancement in tubular heat exchangers using COMSOL Multiphysics and 3D CFD simulations.
- Designed and optimized NACA 0012 inserts with 3, 6, and 9 delta wing protrusions for enhanced thermal performance.
- Developed coupled fluid flow and heat-transfer models using Realizable k- ϵ turbulence and nonisothermal flow.
- Achieved 38.07 W/m²K heat transfer coefficient with 18.03% lower pressure drop in 6 protrusions by geometry optimization.
- Achieved 26.425 W heat transfer with the 6 protrusion design, only 0.052% below the 3 protrusion maximum.
- Authored a conference paper on numerical investigation and optimization of airfoil based heat-transfer enhancement.

Tools and Technologies: COMSOL Multiphysics 6.4, SolidWorks, CAD, CFD, Heat Transfer, Turbulence Modeling, CAD, Thermal Hydraulic Analysis, Nusselt, Prandtl, friction factor, LMTD

PROJECTS

Optimization of Y Shapes NACA Airfoil Inserts for Heat Exchanger:

• *July 2026*

- Designed a Y Shaped NACA 0012 insert with delta wing protrusions for heat transfer enhancement.
- Compared 3, 6, and 9 protrusions using COMSOL, achieved highest heat transfer of 26.4387 W and 38.07 W/m²K with 3 protrusions.
- Selected 6 protrusions for 18.03% lower pressure drop with only 0.052% lower heat transfer.
- **Tools and Technologies:** SolidWorks, COMSOL Multiphysics 6.4, CFD, CAD, Heat Exchanger.

CAD & CFD Analysis of a Swirled Tube Heat Exchanger:

• *July 2026*

- Designed a helically swirled tube heat exchanger in SolidWorks for enhanced thermal performance.
- Simulated 1.2% Al₂O₃-CuO-CNT nanofluid using COMSOL Multiphysics 6.4 and k- ϵ turbulence modeling.
- Achieved 4.14 kW heat duty with 75.3% heat exchanger effectiveness and 220 W/K thermal conductance.
- **Tools and Technologies:** SolidWorks, COMSOL Multiphysics 6.4, CFD, CAD, k- ϵ Turbulence, Heat Transfer.

CFD & Thermal Hydraulic Analysis of a Baffled Shell-and-Tube Heat Exchanger:

• *June 2026*

- Designed 7 tube, 6 baffle shell & tube heat exchanger with alternating segmental baffles for enhanced shell side cross flow.
- Developed and simulated 3D conjugate heat transfer model in COMSOL using k- ϵ turbulence and Nonisothermal Flow.
- Achieved 46.16 kW heat duty with 49.3% effectiveness, heat transfer coefficient of 15,180 W/m²K with 0.68% energy balance error.
- **Tools and Technologies:** SolidWorks, COMSOL Multiphysics, CFD, k- ϵ Turbulence, Thermal-Hydraulic Analysis.

Autonomous Vehicle State Estimation & Optimal Control:

• *June 2026*

- MATLAB autonomous driving framework integrating nonlinear vehicle dynamics, EKF, LQR, and curvature feedforward.
- Implemented a 6 state EKF with 50 Hz IMU + 10 Hz GPS fusion, numerical Jacobian checks, and UKF cross validation.
- Validated sensor fusion across 24 Monte Carlo trials, reducing RMS position error 47% from 0.85 m to 0.45 m.
- **Tools and Technologies:** MATLAB, EKF, UKF, LQR, MPC, Nonlinear Vehicle Dynamics, Monte Carlo Simulation.

High Pressure Hydrogen Refueling Infrastructure Design:

• May 2026

- Designed a 50 km DN300 hydrogen supply pipeline for 4 simultaneous truck fills at 16 kg/min total flow.
- Analyzed turbulent pipeline flow with $Re = 1.16 \times 10^5$, achieving only 0.009 bar pressure drop across 50 km.
- Designed 50→850 bar polytropic compression system, requiring 348.4 kW at 75% efficiency with 17:1 compression ratio.
- Tools and Technologies: Fluid Mechanics, Thermodynamics, Compressible Flow, Nozzle Design. MATLAB.

Design, Analysis & Prototyping of a Static Sector Gate Storm Surge Barrier:

• May 2026

- Designed and modeled cantilevered sector gate in Fusion 360 for a 50 N lateral hydrostatic load using a Warren truss.
- Optimized & simulated the structure and achieved SF 7.2 buckling, SF 20.0 material failure, and 4.5mm max deflection.
- Performed MATLAB calculations, predicted 252 N failure load ($5\times$ design load); withstood 450 N in UTS machine testing.
- Tools and Technologies: Fusion 360, MATLAB, MathCAD, Structural Simulation, UTS Machine.

IoT-Based Smart Home Automation & Safety System:

• April 2026

- Developed ESP32 based IoT platform integrating modules for smart access, gas safety, and automated water management.
- Engineered door locking with <100ms response, achieved 98% accuracy for gas detection and ultrasonic water control.
- Designed the system from breadboard to Proteus simulation and PCB design, reducing ultrasonic noise by 85%.
- Tools and Technologies: ESP32, Arduino/C++, Proteus 9, Vero-board PCB, Sensors, Embedded Systems.

Self Balancing Autonomous Robot with Line Following & Obstacle Avoidance:

• April 2026

- Engineered a 2WD robot with real-time PID balancing, integrated IMU sensing, motor control, and autonomous driving.
- Implemented MPU6050 complementary filter sensor fusion for low latency tilt estimation and closed loop stability control.
- Integrated 3 operating modes: self-balancing, line following, and obstacle avoidance, all governed by the core PID loop.
- Tools and Technologies: C++, Arduino, PID Control, MPU6050 IMU Sensors, IR & Ultrasonic Sensors, Robotics.

Design & Fabrication of a 4WD Autonomous Robotic Vehicle:

• Feb. 2026

- Designed and fabricated a custom 4WD chassis with motorized front-wheel steering for controlled maneuvering.
- Implemented Arduino based navigation with ultrasonic obstacle detection and autonomous avoidance.
- Engineered LED signaling, including front, rear lights, and left/right indicators. mounted solar panel for battery charging
- Tools and Technologies: Arduino, C++. DC 545 Motors, Motor Drivers, Ultrasonic Sensor, Solar Panel, LED's.

Thermodynamic Design & Optimization of a Regenerative Brayton Power Plant:

• Dec. 2025

- Designed a 13.52 MW regenerative Brayton cycle with 2 stage intercooling, 2 stage reheating, and 90% regeneration.
- Analyzed ideal and real cycle performance, achieving 68.9% ideal efficiency versus 12.83% real cycle efficiency.
- Accounted for 79% compressor and 52% turbine efficiencies, quantified an 89.8% drop in work and 886% increase in mass flow.
- Tools and Technologies: EES, Thermodynamics, Brayton Cycle, Regeneration, Intercooling, Reheating, Energy Analysis.

Structural Analysis & FEA Validation of Truss Systems:

• May 2025

- Engineered a 5 joint, 6 member truss and evaluated its response to a 300 N horizontal load using the Method of Joints.
- Simulated the complete truss in ANSYS Workbench, reproducing the ± 212.13 N maximum axial forces.
- Verified static equilibrium through 300 N horizontal and ± 150 N vertical support reactions.
- Tools and Technologies: ANSYS Workbench, FEA, Engineering Mechanics, FEA, Applied Mechanics.

Precision CAD Design & Assembly of Stepped Shaft:

• Dec. 2024

- Engineered a high fidelity stepped shaft and support bracket assembly in AutoCAD with production ready geometry.
- Achieved 99.8% model accuracy and 100% tolerance compliance through verification and assembly fit validation.
- Translated complex orthographic drawings into precise 3D CAD models, stepped sections, flange, bore, and D-cut features.
- Tools and Technologies: AutoCAD, Geometric Dimensioning and Tolerancing (GD&T), Manufacturing Engineering.

Relevant Coursework:

- Core Engineering:** Thermodynamics (2 Course Series), Fluid Mechanics (2 Course Series), Heat & Mass Transfer, Engineering Mechanics (2 Course Series), Mechanics of Materials (2 Course Series).
- Design & Materials:** Computer Aided Drawing, Engineering Drawing & Graphics, Numerical Analysis, Manufacturing Process, Material Engineering, Engineering Dynamics. HP 3D Printing Course.
- Systems & Programming:** Computer Systems & Programming, Electrical & Electronics Engineering.

TOOLS & SKILLS:

Tools & Simulations: SolidWorks, AutoCAD, COMSOL Multiphysics ANSYS, CFD, OpenBom, EES, Primavera P6, Proteus 9.

Thermal & Fluid Analysis: Heat Transfer, k-ε Turbulence, Nonisothermal Flow, LMTD, ε-NTU, Compressible Flow.

Developer Tools: Git & GitHub, VS Code, Jupyter Notebook, Arduino IDE.

Programming Languages: Python, C/C++, HTML, MATLAB, Maplesoft,

Microcontrollers: ESP32, STM32, ATmega 328p, Arduino.

Soft Skills: Project Management, Teamwork & Leadership, Complex Problem Solving, Technical Report Writing.